SCHOOL BUS ROUTE OPTIMIZATION

Comparing Nearest Neighbor, Branch and Bound, and Brute Force



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1. What problem are we solving?

- A bus has to pick up students from three intermediate locations.
- The trip starts at the selected city and ends at the school.
- The route should cover every pickup location once.
- The bus has a fixed capacity of 50 students.

Main objective

Find a shorter order for visiting the pickup locations before reaching the school.

2. Data used in the program

Rajahmundry

Pickup points: Lalacheruvu, Danavaipeta, Gandhipuram 1st Street
Students: 12 + 10 + 13 = 35

Peddapuram

Pickup points: Government Hospital, Bus Stand, Road Area
Students: 10 + 12 + 13 = 35

Samalkot

Pickup points: Mehar Complex, Gandhi Statue Center, APSRTC Bus Stand
Students: 12 + 10 + 13 = 35

Bus capacity = 50 students | Total students in each city = 35

3. How the route is represented



- The distance matrix stores the distance between every pair of locations.
- The route distance is the sum of the distances between consecutive locations.
- The final leg from the last pickup point to the school is included.

4. Algorithms used

Nearest Neighbor (Greedy)

Starts at the city and repeatedly chooses the closest unvisited pickup point. It is quick, but it may not give the shortest overall route.

Branch and Bound

Tries possible pickup orders. It stops exploring a partial route when its current cost is already no better than the best complete route found.

Brute Force

Checks all possible orders of the intermediate pickup points. It is used as a verification method for the minimum distance.

In this code, the start and end locations stay fixed; only the three middle stops are rearranged.

5. Results from the given distance data

City	Nearest Neighbor	Branch & Bound	Distance saved
Rajahmundry	72 km	70 km	2 km
Peddapuram	44 km	40 km	4 km
Samalkot	27 km	26 km	1 km

Branch and Bound matches Brute Force for all three cities.

This means the optimal distances are verified by the exhaustive search.

6. Best routes found

Rajahmundry

70 km

Rajahmundry → Lalacheruvu → Danavaipeta → Gandhipuram 1st Street → School

Peddapuram

40 km

Peddapuram → Government Hospital → Bus Stand → Road Area → School

Samalkot

26 km

Samalkot → Mehar Complex → Gandhi Statue Center → APSRTC Bus Stand → School

7. Verification and limitations

Verification

The program compares Branch and Bound with Brute Force. Matching minimum distances indicate that the Branch and Bound result is correct for these inputs.

Limitations

Only three pickup locations are included per city. The program handles one bus and one trip, with a fixed capacity of 50.

- Nearest Neighbor is fast, but it does not guarantee the shortest route.
- Brute Force becomes slow as the number of pickup points increases.
- A future version could accept live locations, more stops and multiple buses.

8. Conclusion

The project shows how algorithms can help plan a school bus trip with less travel distance.

- Branch and Bound found the shortest route for each of the three city datasets.
- Brute Force confirmed the same minimum distances.
- Nearest Neighbor gave a useful quick result, but its routes were slightly longer.

Thank You!